

EXECUTIVE SUMMARY

UAIL IT Dashboard

Operational wallboard and admin control surface for UAIL IT operations, combining live infrastructure, service-desk, and network telemetry into a single managed web stack.

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USER SURFACES

2

CORE DATA DOMAINS

4

PRIMARY RUNTIME STORE

SQLite

DEPLOYMENT TARGET

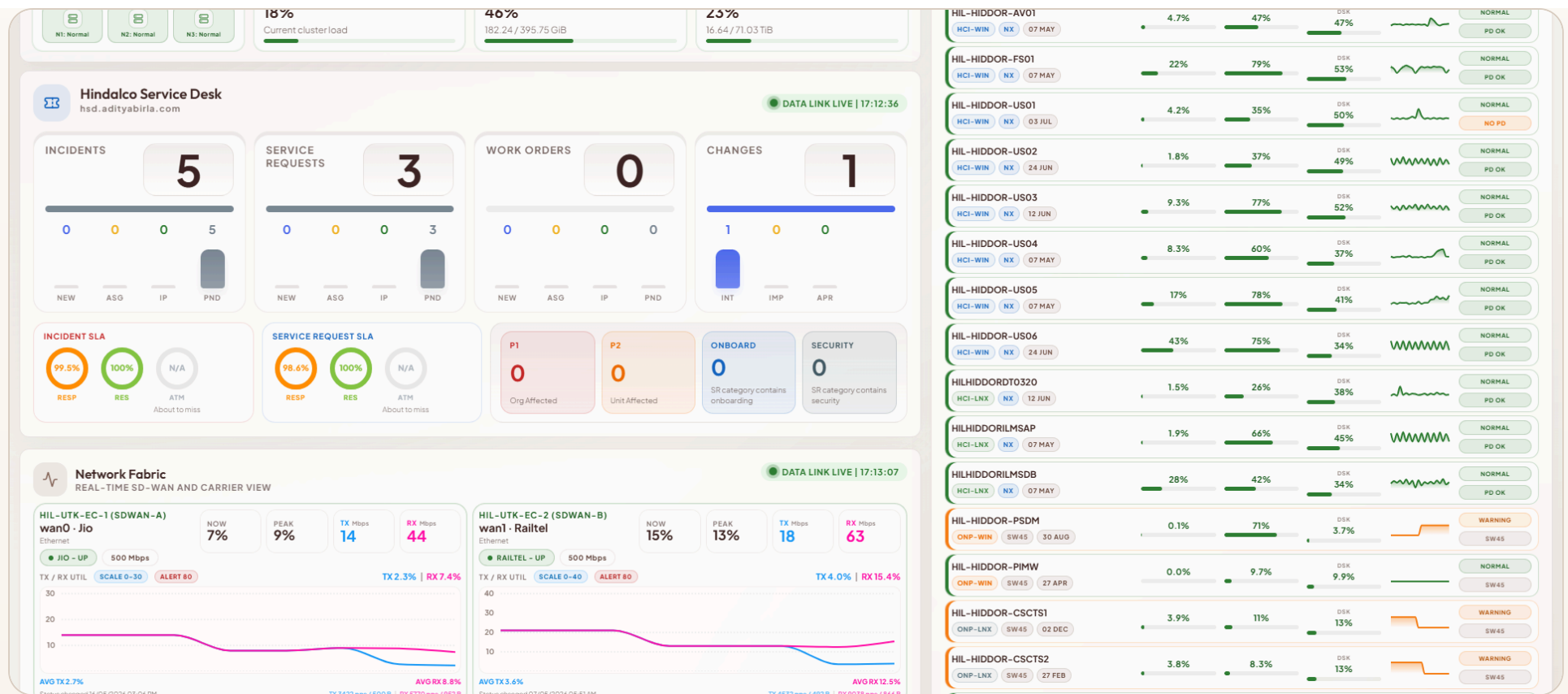
Windows Server

WHAT THE PRODUCT DELIVERS

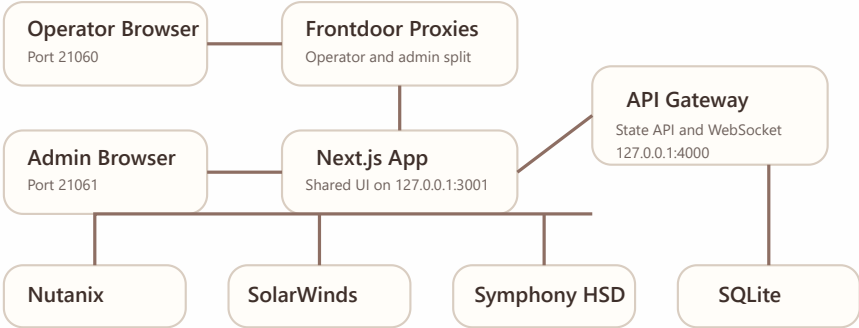
- One-screen engineering dashboard for HCI, HSD, network, and servers.
- Separate operator and admin web surfaces using the same deployed application stack.
- Live section freshness, last-sync visibility, and source-attribution rules for trustable operations.
- Admin controls for service restart, source configuration, and collector session recovery.

KEY OPERATING PRINCIPLE

The dashboard must not fabricate values during normal operation. If a collector fails, the system keeps showing the last successfully synced data together with the last synced time for the affected card or section.



Single web stack with controlled separation



CORE DESIGN CHOICES

- Two exposed LAN surfaces only: operator and admin.
- Internal UI and gateway remain on loopback.
- SQLite stays primary for runtime simplicity.
- PostgreSQL remains optional for mirroring, config, and secret storage.
- PM2 manages the full multi-process runtime on Windows.

COLLECTORS

Nutanix	Direct REST API
SolarWinds	Headless browser
Symphony HSD	Headless browser

REFRESH CADENCE

HCI	30 sec
Servers/Network	30 sec
HSD	60 sec

PRIMARY UX TARGET

One-page engineering wallboard at 1920x1200, with a responsive stacked layout for tablet and mobile access.

Source-of-truth rules and freshness behavior

SOURCE RULES

- Nutanix is the source of truth for HCI-backed servers.
- SolarWinds is the source of truth for network telemetry and on-prem server telemetry.
- For overlapping server coverage, Nutanix remains primary unless it is stale for more than 10 minutes and SolarWinds telemetry exists.

FRESHNESS RULES

- Each section stores last attempt, last success, poll interval, and last error.
- Cards show the last known good data when collectors fail.
- The UI uses section health states instead of synthetic fallback values.

WHAT OPERATORS CAN TRUST

- Displayed values come from real collector output, not fabricated placeholders.
- Data-link status tells whether the values are current, stale, errored, or never synced.
- Fallback state is explicit when Nutanix-backed servers temporarily show SolarWinds values.

IMPLICATION FOR REVIEW

This is a status-preserving operational dashboard, not a predictive analytics surface. Review and audit discussions should focus on freshness, source attribution, and recovery controls rather than whether missing data is synthesized.

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2/2

4/4

Overview

Stack health and quick actions

Services

Run state and restart controls

Sessions

Imports and reauthentication

Sources

Collector endpoints and credentials

Sessions

Import copies refreshed storage-state JSON. Reauthentication opens an interactive browser on the server host.

HSD Session

Saved session is authenticated against the live portal.

AUTH OK

Primary session file | Updated 2026-07-17T11:44:21.350Z | 16.4 KB | Authenticated against the live HSD dashboard. | Checked 2026-07-17T11:45:08.007Z Import Session Launch Reauth on Server Import Legacy HSD Profile**SolarWinds Session**

Saved session is authenticated against the live portal.

AUTH OK

Servers portal session | Updated 2026-07-17T11:44:55.255Z | 9.5 KB | Authenticated against the live SolarWinds portal. | Checked 2026-07-17T11:45:06.295Z**Networks portal session** | Updated 2026-07-17T11:44:51.878Z | 9.8 KB | Authenticated against the live SolarWinds portal. | Checked 2026-07-17T11:45:06.354Z Import Session Launch Reauth on Server

Pragmatic internal deployment with compensating controls

NEEDS DOCUMENTED
CONTROLS

CONTROLS ALREADY PRESENT

- Separate operator and admin surfaces.
- Signed role cookies and admin-only API guards.
- Loopback-only runtime config and secret endpoints.
- Optional encrypted secret storage in PostgreSQL.

KNOWN LIMITATIONS

- Local password-only auth today.
- Browser session files required for SolarWinds and HSD.
- Internal HTTP still exists inside the runtime model.
- Nutanix TLS verification is relaxed due to source certificate issues.

COMPENSATING CONTROLS

- Deploy on a dedicated internal Windows server or VM.
- Expose only ports 21060 and 21061 to the LAN.
- Restrict access by firewall and network ACL.
- Lock runtime folders to admins only.

RECOMMENDED NEAR-TERM HARDENING

- Move production secrets into PostgreSQL encrypted storage.
- Rotate `APP\_AUTH\_SECRET` and the shared local password at deployment time.
- Introduce stronger application-layer authentication for administrative writes.
- Put the front doors behind internal TLS termination when feasible.

AUDIT POSTURE

The application is defensible for an internal ISMS review when positioned as a controlled, LAN-hosted operational dashboard on a dedicated server with documented compensating controls and operational procedures.

Simple Windows-hosted operational model

DEPLOYMENT PATTERN

- Install on Windows Server 2019 or later.
- Use the packaged installer or offline bundle.
- Run the stack under PM2.
- Use the admin surface for service status, restart, and session management.

RUNTIME ENDPOINTS

Operator surface	21060
Admin surface	21061
Shared UI	127.0.0.1:3001
Gateway	127.0.0.1:4000

ADMIN CAPABILITIES

- Service status and restart controls
- Collector source configuration
- HSD and SolarWinds session validation
- Server-local HSD reauthentication

OFFLINE READINESS

- Installer artifact built
- Offline server bundle built
- Artifact hashes generated
- Deployment runbooks retained under `deployment/`

SUPPORT MODEL

- Operator use is browser-only
- Admin use is browser-first
- No separate desktop admin app required
- Session recovery remains an admin operation

Control Overview

Use this page as the first stop for stack health, quick recovery, and source readiness.

LIVE

SERVICES NEEDING ATTENTION

0

[Open services](#)

SESSIONS NEEDING RENEWAL

0

[Open sessions](#)

ENABLED SOURCE TARGETS

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[Open sources](#)

[Open Operator View](#)

[Refresh Console](#)

[Restart Entire Stack](#)

What Changed

Each workflow now sits in its own tab so service actions, session recovery, and source editing stay separated.

Services

Run state and restart controls

[Open](#)

Sessions

Imports and reauthentication

[Open](#)

Sources

Collector endpoints and credentials

[Open](#)

Service Snapshot

Live PM2 and gateway status across the shared dashboard stack.

API Gateway

HTTP 200

ONLINE

Dashboard UI

HTTP 200

ONLINE

Session Snapshot

Imported browser state and renew / reauthenticate readiness.

Immediate next actions after documentation baseline

PRIORITY 1

- Approve the current architecture baseline and controlled deployment model.
- Deploy on a dedicated Windows server with LAN restriction and hardened filesystem permissions.
- Move operational secrets to PostgreSQL encrypted storage for production.

PRIORITY 2

- Complete deeper audit and redesign work for the server data model if required.
- Reduce dependence on browser-authenticated HSD access where enterprise API options become available.
- Introduce stronger auditability and authentication for administrative actions.

SUMMARY STATEMENT

The UAIL IT Dashboard is now in a materially stronger state: the UI is operationally useful, the runtime model is simpler, the trust model is explicit, the deployment path is defined, and the information-security position is documentable. The remaining work is primarily hardening and control maturity rather than uncertainty about what the product is or how it operates.

CURRENT BASELINE

Documented

OPERATING MODEL

Defined

NEXT FOCUS

Hardening

